

STATUS REPORT

Platform, infrastructure, architecture & UX

Top-priority problem list with risk & urgency

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AUTHOR

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Executive Summary

Where NutriTrack stands today, in 4 numbers

30%

Production-ready

Code is structured
but doesn't build

9,373

Dart LOC

across 6 features,
clean architecture

31

Compile errors

block next build,
all in 2 files

0

Tests

zero coverage,
test/ doesn't exist

WHAT'S WORKING

✓ AI gateway architecture

✓ Drift DB schema (7 tables)

✓ PB live (8 nt_* collections)

HALF-BUILT

⚠ 5/6 screens render fake data

⚠ No auth / onboarding route

⚠ Workout session UI missing

BLOCKERS

✗ App won't compile

✗ Never run on a device

✗ 9 packages on major N-1

Platform Status

Flutter SDK, dependencies, platform readiness

STACK SNAPSHOT

Flutter	3.27+ (declared) · 3.29.3 in your lockfile	✓
Dart	3.6+ · supports records, patterns, sealed	✓
State	Riverpod 2.6.1 with @riverpod codegen	✓
Routing	go_router 14.8.1 (StatefulShellRoute)	✓
Local DB	Drift 2.28.2 (7 tables, all defined)	✓
AI Gateway	OpenRouter → MiniMax M3 + Gemini + GPT-4o	✓
Voice	record 5.1.2 → Whisper transcription	✓
Barcode	mobile_scanner 5.2.3 + OFF client	⚠
Camera	camera 0.11.4 → 0.12.0+1 available	⚠
Cloud sync	Supabase declared but unused	⚠
Health	REMOVED in this sprint (incompat)	✗

BUILD READINESS

Android compile	✗ FAILS
Android (post-fix)	STAGING
iOS compile	UNKNOWN
Device run	NEVER
flutter analyze	31 errors + 17 warnings + 62 info
dart fix --apply	WOULD KILL 12 of 17 warnings
pubspec.lock	GITIGNORED
Generated files	GITIGNORED
Secrets config	TEMPLATE ONLY

Infrastructure Status

PocketBase, GitHub Actions, dependencies

POCKETBASE · LIVE SCHEMA

9 collections on pocketbase.scaleupcrm.com

nt_users	0 records	auth-anchored profiles
nt_food_logs	0 records	synced meal entries
nt_exercises	46 records	chest-only seeded
nt_workout_sessions	0 records	session history
nt_weight_entries	0 records	weight log
nt_favorites	0 records	starred foods
nt_meal_templates	0 records	recurring meals
nt_sync_queue	0 records	offline→cloud queue
nt_barcode_cache	0 records	OFF cache (anon read)

GITHUB ACTIONS · CI/CD

3 workflows + 1 README

- ci.yml
ubuntu/macos runners · analyze + test + build APK/iOS
- off-to-pb-sync.yml
cron 0 */6h · fetches OFF products → PB cache
- dependency-sync.yml
weekly · opens PR for outdated packages

SECRETS

- PB_URL in workflow env
- PB_ADMIN_EMAIL in workflow secrets
- PB_ADMIN_PASSWORD in workflow secrets
- OPENROUTER_API_KEY in secrets.dart (template only)

Architecture

Domain-Driven Design · 6 features · 7 Drift tables

FEATURES BY SIZE

camera	snap → AI → review	1962
dashboard	today's meals + macros	1969
barcode	scan → OFF lookup	1654
workout	search + log	1046
settings	TDEE + profile	736
insights	weight + trends	481

DRIFT TABLES (7)

FoodLogEntries	meal entries + macros + source
ExerciseEntries	logged exercises per session
WorkoutSessions	active + history sessions
WeightEntries	weight log + date
UserProfiles	TDEE + goal + biometrics
ImageHashCache	AI dedup (avoid re-recognize)
PendingSyncEntries	offline → cloud queue

DATA FLOW

Capture → AI Gateway → Drift (local) → PocketBase (cloud) · 3-tier barcode cache: in-memory → PB → OFF (anonymous read)

Current Task Status

What's done, what's in progress, what's blocked

DONE12

✓

Drift migration (Isar removed)

✓

PB schema live (9 collections)

✓

46 chest exercises seeded

✓

AI gateway architecture

✓

Drift codegen working

✓

Brand identity (icon, palette)

✓

Brand exports (iOS/Android/Web)

✓

KNOWN_WARNINGS.md written

✓

Gradle 8.14.0 + AGP 8.11.1

✓

record 5.1.2 pin (Linux iface)

✓

health plugin removed

✓

media/ golden-source structure

IN PROGRESS3

🔄

Audit doc v2 (PRODUCTION_READINESS)

🔄

Strategy doc (WHAT_MISSING)

🔄

Major-version debt tracking (\$6)

BLOCKED5

✗

Android compile
31 errors in 2 files

✗

First device run
no working APK/IPA

✗

Insights screen
renders fake sample data

✗

Workout session UI
no active-session flow

✗

Onboarding flow
no /onboarding route

UX Status

What users would see today if the app compiled

SCREENS

6

dashboard, workout,
insights, settings,
camera, barcode

WIRED TO DATA

1 of 6

5 of 6 screens render
placeholder / sample data

EMPTY STATES

0

no `shimmer`, no
empty-state illustrations

MOTION + HAPTIC

minimal

app_motion exists
but underused

SCREEN-BY-SCREEN UX READINESS

Dashboard	✓ real data wiring	✗ fake data, no onboarding
Workout	✓ search working	✗ no session UI, no rest timer
Insights	✓ fl_chart present	✗ renders `_generateSampleWeights()`
Settings	✓ TDEE wizard OK	✗ no first-launch gate
Camera	✓ build broken	✗ 31 compile errors
Barcode	✓ build broken	✗ RRect.topLeft undefined

Performance Suggestions

What to fix before users notice the lag

1	Skeleton loaders on every list Dashboard tiles, exercise search, history, insights charts all flash empty→full. Add shimmer widgets.	~4 hours	BIG
2	RepaintBoundary around MacroDonut Already added in earlier commit. Verify on real device — fl_chart redraws on every frame otherwise.	~1 hour	MED
3	Cache Network images (cached_network_image) Barcode images already wrapped. Apply same pattern to food images from OFF / PB.	~2 hours	MED
4	Lazy-load PB collections Currently FoodLogRepository fetches full day on every screen mount. Use Drift streams (watchByDate) instead of getByDate.	~3 hours	BIG
5	Profile-mode build for production flutter build apk --release --obfuscate --split-debug-info=debug/. Cuts ~30% APK size.	~30 min	QUICK
6	Image compression before AI Camera capture = 2-4 MB JPEG. Resize to 1024px, JPEG quality 85 = ~200 KB. Faster AI upload, lower bandwidth.	~2 hours	MED
7	Skia caching for static content macro_donut, wordmark, empty states — wrap in RepaintBoundary or const where possible.	~2 hours	MED
8	Sync TextEditingController (already done) Workout search field — verify no sync creation in build(). You fixed this already. ✓	✓	DONE

Security Suggestions

What to lock down before the first user logs in

API keys not in code	OK	OpenRouter key in secrets.dart (template-only today). Never committed.
Secrets.dart in .gitignore	OK	.gitignore line 47: lib/core/config/secrets.dart. Verified.
PB admin creds in CI	OK	GitHub Actions secrets only. Not in any yml file. Verified.
Barcode cache: anonymous read	OK	PB listRule="" → world-readable. Only OFF data, no PII.
Crash reporting	MISSING	No Sentry, no Crashlytics. PII may end up in error logs unfiltered.
TLS pinning	MISSING	Dio talks to PB and OpenRouter over HTTPS, but cert pinning not configured. Acceptable for v1; revisit for v2.
Image storage on device	PARTIAL	Camera captures go to app-private dir (good). But they're never deleted; could leak via forensic tools.
PII in error logs	RISK	AppErrorHandler captures PlatformDispatcher.onError — should redact email, weight, food names before logging.
Supabase auth	MISSING	Auth is currently local-only. Once you re-add Supabase, must configure RLS on every table from day 1.
Dependency CVE scan	MISSING	No automated npm/advisory scan. GitHub Dependabot could enable in 5 minutes.
ProGuard / R8 obfuscation	ENABLED	android/app/build.gradle: minifyEnabled true, shrinkResources true. ✓
Debug signing for release	RISK	build.gradle uses the debug keystore for release builds. Acceptable for early access; MUST replace before public launch.

UX Suggestions

What separates "functional app" from "delightful app"

Onboarding	MISSING	3-screen first-run: welcome → why-we're-different → TDEE wizard. Without it users churn in 30s.	2 days	CRITICAL
Empty states	MISSING	Every screen needs a 'nothing here yet' moment with personality + a CTA. 8 illustrations needed.	1 day	HIGH
Haptic feedback	PARTIAL	HapticFeedback.mediumImpact on meal log, workout complete, set save. iOS-native feel.	2 hours	MED
Sound design	MISSING	Swoosh on log, bell on workout complete. just_audio already in pubspec, just unused.	4 hours	MED
Streaks + badges	MISSING	5-day logging streak, 7-day protein goal hit. Duolingo's #1 retention mechanism. Pure motivation.	3 days	HIGH
Pull-to-refresh	MISSING	Dashboard re-aggregates. Insights re-queries. Standard mobile pattern.	2 hours	LOW
Quick actions	MISSING	Long-press dashboard tile → quick log. Swipe-to-delete meal. iOS share extension later.	1 day	MED
Offline indicator	MISSING	Banner: 'Offline — logging locally, will sync when reconnected' + queue count.	3 hours	MED
Wow moment	MISSING	First time user logs via voice or camera → confetti + 'Saved 2 minutes vs typing!'	3 hours	HIGH
Error messages	PARTIAL	AppErrorHandler exists but messages should be human, specific, actionable.	1 day	MED
a11y	MISSING	No Semantics audit, no contrast check, no dynamic font scaling tested.	2 days	MED
i18n	MISSING	English-only is fine for v1, but structure (.arb) must support it. ~150 strings to externalize.	2-3 days	LOW

Top Problems · Critical · Act NOW

Blockers that prevent any further work

01

Build is broken — 31 compile errors

WHERE: lib/features/barcode/presentation/screens/barcode_scanner_screen.dart (27 errors) +
lib/features/camera/presentation/screens/camera_screen.dart (4 errors)

WHAT: RRect.topLeft/topRight/bottomLeft/bottomRight undefined (use Rect or RRect.tlRadius). 3 callbacks with
`Function()` instead of `Function(String)`. MobileScannerErrorBuilder API drift.

IMPACT: App won't run on any device. Nothing else matters until this is fixed.

EFFORT

45 min

TASK #1: Mechanical patch

02

Zero tests, no test/ directory

WHERE: No unit tests, no widget tests, no integration tests.

WHAT: flutter test --coverage in CI passes only because there are no tests to fail. Every future refactor is a coin flip.

IMPACT: Regressions slip through silently. New contributors can't find patterns to copy.

EFFORT

1-2 days

TASK #5 in roadmap

03

No onboarding, no first-run experience

WHERE: Initial route is /dashboard. No /login, /signup, or /onboarding route exists.

WHAT: User opens the app, sees generic 2000 kcal target with no context. Churns in 30 seconds.

IMPACT: Without onboarding, retention is impossible to measure or improve.

EFFORT

2 days

TASK #6 in roadmap

04

All data wiring is fake / sample

WHERE: Insights screen renders `\_generateSampleWeights()`. Workout screen has no session UI. Dashboard donut takes fake
numbers. Camera/barcode review sheets show static UI.

WHAT: 5 of 6 main screens are basically tech demos. Drift tables exist but are unused.

IMPACT: Without real data, the app cannot ship. No amount of polish hides this.

EFFORT

4 days

TASK #7 in roadmap

Top Problems · High Priority · This Week

After the critical path, address these

05	No crash reporting (Sentry/Crashlytics) No sentry_flutter, no firebase_crashlytics dep. AppExceptionHandler captures errors but doesn't report them anywhere. IMPACT: Production without crash reporting = blind. You can't fix bugs you don't know about. PII risk if errors aren't redacted before logging.	EFFORT 2 hours	FIX Add sentry_flutter, wrap AppExceptionHandler
06	Voice logging UI half-built record: ^5.1.2 + Whisper path defined in AI gateway. Quick log bar has a mic icon. But no hold-to-talk button, no waveform animation, no transcription feedback. IMPACT: This is NutriTrack's #1 differentiator vs MyFitnessPal. Without it, app is just another calorie tracker.	EFFORT 3 days	FIX Hold-to-talk + waveform + Whisper wiring
07	AI camera lacks verification flow Camera captures, AI returns FoodLogEntry list with confidence. But no UI to show confidence bars or ask user to confirm low-confidence items. IMPACT: Without verification, users lose trust the moment AI guesses wrong. Need red/yellow/green confidence indicator + 'Take another photo?' escape hatch.	EFFORT 3 days	FIX Add confidence UI + quick-fix shortcuts
08	Exercise DB: 46 of 488 scripts/exercises_chest.js is the only seed file. Back/shoulders/arms/legs/glutes/core/cardio/full-body missing. IMPACT: Workout feature is half-built. User can search but most muscle groups show zero results.	EFFORT 4-6 hours	FIX Boilerplate: 8 more .js files
09	Workout session UI missing WorkoutScreen has search + list. But no start session → log sets → rest timer → end session flow. No active-session screen, no PR tracking. IMPACT: Half of NutriTrack's value prop is the workout side. Without session UI, it's a search tool, not a tracker.	EFFORT 4 days	FIX Active session + set logger + rest timer

Top Problems · Medium Priority · This Month

Polish, performance, security hardening

10	9 packages on major N-1 flutter_riverpod 2.6 (latest 3.3), freezed 2.5 (3.2), mobile_scanner 5.2 (7.2), health REMOVED, go_router 14.8 (17.3), intl 0.19 (0.20), etc. IMPACT: Future Flutter versions break these. Each is a real migration: API renames, codegen output changes, permission flow drift.	EFFORT 1 week	FIX 5 grouped PRs per KNOWN_WARNINGS \$6
11	7 dead dependencies in pubspec just_audio, web_socket_channel, confetti, flutter_staggered_animations, animations, lottie, gap, shimmer, supabase_flutter. IMPACT: 0 imports in lib/ for most. ~10 MB wasted APK size. Slower builds.	EFFORT 30 min	FIX flutter pub remove for each
12	README lies about Isar Stack table says Local DB Isar 3.1. You migrated to Drift 5 commits ago. IMPACT: Confuses every new contributor. Costs an hour of confusion per onboarding.	EFFORT 10 min	FIX 3-line edit + bump version
13	Release build uses debug keystore android/app/build.gradle: signingConfigs.release = signingConfigs.debug IMPACT: Acceptable for internal testing. CRITICAL to replace before public launch — Play Store / App Store will reject.	EFFORT 1 hour	FIX Generate release keystore, store in key.properties
14	Adaptive TDEE engine not built TDEE wizard calculates starting value. But no weekly recalibration from real weight + intake data. IMPACT: MacroFactor's killer feature. Without it, NutriTrack's target is static even after 6 months of logging.	EFFORT 1 day	FIX 7700 kcal/kg formula + weekly trigger
15	Image storage on device Camera captures saved to app-private dir. Never deleted. IMPACT: Old food photos pile up. Forensic tools can recover. PII risk.	EFFORT 2 hours	FIX Retention policy + cleanup cron

WHAT TO DO NEXT

Three steps to get NutriTrack shipping.

TODAY

Fix the build.

Run the 5-line edit list for RRect + Function() signatures. 45 minutes. Open the icon. Run flutter run.

THIS WEEK

Wire the data + add Sentry.

Replace 5 sample-data screens with real Drift. Add Sentry (2 hours). Wire 7 priority tests. ~3 days.

THIS MONTH

Ship differentiators.

Voice UI + camera verification + workout session + complete exercise DB. ~3 weeks. Then TestFlight + public beta.

Repo: github.com/albertlaudia/nutritrack · PB: pocketbase.scaleupcrm.com · 9,373 Dart LOC